Characterizing Fairness Trade-offs in Multi-Client Adaptive Streaming

***Abstract*—Ensuring fairness among users competing for shared network resources remains a critical challenge in multi-client adaptive video streaming. While modern adaptation strategies optimize individual Quality of Experience (QoE), they often overlook the complex trade-offs between different fairness dimensions and playback stability. This study presents an analytical characterization of these trade-offs by evaluating six diverse adaptation strategies—ranging from heuristic-based baselines to fairness-aware reinforcement learning—under identical, highly variable network conditions. We employ a multi-metric framework that jointly analyzes four complementary fairness indicators: Time-Fair QoE (TF-QoE), QoE Fairness Score (QFS), Bitrate Fairness Index (BFI), and Stability-Aware Fairness (SAF). By analyzing experimental data from a 10-client edge-assisted testbed replaying real-world 4G/3G traces, we reveal structural conflicts between quality equity and adaptation instability. Our findings demonstrate that gains in bitrate fairness often come at the expense of increased quality switches for specific users, a trade-off that is significantly amplified under constrained network conditions. This work provides a rigorous observational baseline for the design of equitable streaming systems, highlighting that effective fairness evaluation must account for the interplay between resource distribution and temporal stability.**

***Index Terms*—Adaptive Bitrate Streaming, Fairness Trade-offs, Multi-Client Video Streaming, QoE Evaluation, Network Variability**

I. Introduction

The evolution of edge-assisted video delivery has transformed Adaptive Bitrate (ABR) streaming into a complex multi-agent resource contention problem. While modern adaptation logic effectively optimizes individual Quality of Experience (QoE), the structural interactions between concurrent clients sharing constrained resources remain insufficiently characterized. Traditional fairness evaluation relies heavily on aggregate indices like Jain's Fairness Index (JFI), which typically ignore the temporal and stability-related disparities that define modern user experience.

This study deconstructs these disparities through a multi-metric observational analysis across four dimensions: temporal quality distribution (TF-QoE), distributional fairness (QFS), bitrate allocation (BFI), and adaptation stability (SAF). By evaluating six distinct ABR strategies under real-world 4G/3G network traces, we expose the "fairness-stability trade-off"—a structural conflict where gains in resource equity often drive increased playback instability for a subset of the client fleet. Our findings indicate that network variability acts as an amplifier for these conflicts, revealing that systems appearing "fair" under stable conditions frequently exhibit severe temporal unfairness during transient congestion windows.

The main contributions of this work are: (i) the identification of structural conflicts between bitrate equity and inter-client stability variance; (ii) the characterization of the "tactical panic" and "recovery lag" phenomena as drivers of temporal unfairness; and (iii) a set of design recommendations for the transition toward stability-aware, multi-metric fairness controllers in edge-assisted environments.

The remainder of the paper is organized as follows. Section II provides the conceptual grounding. Section III reviews the literature. Section IV describes the evaluation methodology. Section IV-D presents the analytical results. Finally, Section V concludes the paper.

II. Background

Adaptive Bitrate (ABR) streaming enables video clients to dynamically select video representations according to observed network conditions and playback state. By adapting quality over time, ABR mechanisms aim to mitigate rebuffering events, reduce startup delays, and limit disruptive quality oscillations under fluctuating bandwidth conditions. DASH-based implementations, such as DASH.js, are widely adopted in both academic research and industrial deployments, serving as representative baselines for adaptive streaming behavior. Classical ABR strategies rely on heuristic decision rules derived from throughput estimation or buffer occupancy control [1]–[3]. While such approaches can achieve satisfactory performance in single-client scenarios, their behavior becomes significantly more complex in multi-client environments where users compete for shared and constrained network resources. Independent adaptation decisions may lead to uneven quality allocation, increased playback instability, or prolonged low-quality exposure for some users, even when overall network utilization remains efficient. Learning-based approaches, particularly those based on reinforcement learning (RL), have been explored as alternatives to heuristic ABR control. Systems such as Pensieve [4] demonstrate that data-driven policies can capture complex adaptation dynamics and improve individual Quality of Experience (QoE). However, these approaches remain primarily QoE-centric and typically evaluate fairness only as a secondary or post-hoc metric. Moreover, commonly adopted indicators such as Jain's Fairness Index (JFI) [5] provide a limited view of fairness, as they do not account for temporal dynamics, instability effects, or prolonged exposure to degraded quality. Recent studies have emphasized that fairness in adaptive streaming cannot be adequately characterized by a single metric. Complementary fairness indicators have been proposed to capture different

dimensions of user experience, including distributional equity, temporal exposure, throughput allocation, and playback stability [6]–[8]. Taken together, these metrics reveal that fairness is inherently multi-dimensional and may conflict with traditional QoE objectives, motivating a broader evaluation perspective when analyzing multi-client adaptive streaming systems under heterogeneous network conditions.

III. Related Works

While prior research has established robust foundations for QoE optimization [3], [4], the interaction between fairness and stability in multi-client scenarios remains insufficiently characterized. This study complements existing works by providing the specific analytical evidence needed to transition from single-metric fairness to multi-dimensional, stability-aware controllers. Previous works by the authors have explored QoE optimization and fairness evaluation in adaptive streaming systems [9], [10], providing the experimental basis upon which the present analytical study is conducted. Research on adaptive video streaming has extensively investigated mechanisms for improving Quality of Experience (QoE), stability, and resource utilization in heterogeneous network environments. Early work on Adaptive Bitrate (ABR) streaming focused on heuristic strategies based on throughput estimation and buffer occupancy control. Representative approaches such as FESTIVE [1] and BOLA [3] improved playback stability and efficiency under variable network conditions. However, these methods rely on independent client-side decisions and do not explicitly address fairness among multiple users competing for shared bandwidth, often resulting in uneven quality distribution or instability in congested scenarios. Reinforcement learning (RL) has been explored as an alternative paradigm for ABR control, aiming to capture complex adaptation dynamics through data-driven policies. Pensieve [4] demonstrated the feasibility of learning bitrate selection strategies directly from interaction trajectories, inspiring subsequent extensions that focus on model tuning, efficiency, or content awareness, such as Oboe [11] and HotDASH [12]. While these approaches improve individual QoE and robustness under fluctuating conditions, fairness is typically considered only as an evaluation metric or is not explicitly addressed during decision-making, limiting their applicability in multi-client environments. In parallel, several studies have introduced fairness-aware mechanisms for adaptive streaming. Metrics such as the QoE Fairness Score (QFS) [7] and the Bitrate Fairness Index (BFI) [13] propose alternative fairness formulations by capturing distributional aspects of user experience and bitrate allocation, respectively. Other works investigate distributed or optimization-based bandwidth allocation policies to mitigate unfair resource distribution at network bottlenecks [14]. Multi-agent learning approaches, such as MuABR [15], further explore fairness-aware adaptation by coordinating decisions across clients. Despite these advances, fairness is typically modeled using a limited set of metrics or evaluated under specific scenarios, without explicitly analyzing trade-offs between different fairness dimensions or their interaction with QoE stability. Edge-assisted adaptive streaming has also gained attention as a means to improve responsiveness and scalability. Systems such as ARARAT [16] and other edge-based solutions [17], [18] investigate collaborative mechanisms for QoE optimization and load balancing. More recent works examine fairness-aware resource allocation in edge and MEC environments [19], [20]. Nevertheless, most of these studies either rely on aggregate fairness indicators, such as Jain's Fairness Index, or treat fairness as a secondary objective, providing limited insight into how different fairness metrics interact with QoE stability and adaptation behavior.

Overall, existing literature has made substantial progress in ABR optimization, fairness-aware evaluation, and edge-assisted streaming architectures. However, prior work largely focuses on improving specific performance objectives or incorporating fairness as a single aggregated criterion. Explicit analyses of trade-offs between multiple fairness metrics and their interaction with QoE stability and adaptation dynamics remain scarce. Table I summarizes representative studies from the literature, highlighting the diversity of fairness perspectives adopted and the limited attention given to multi-metric trade-off analysis. This gap motivates the present work, which complements existing studies by systematically analyzing fairness trade-offs in multi-client adaptive streaming systems using a multi-metric evaluation perspective.

IV. Trade-off Analysis Methodology

The experimental data analyzed in this study are derived from controlled executions previously reported in related works [9], [10] and are reused exclusively for post-execution analysis. The analysis builds upon a validated experimental framework previously introduced by the authors [9], [10], enabling a focused investigation of fairness trade-offs without confounding architectural variations. By maintaining consistent architectural assumptions across experiments, the analysis isolates the impact of adaptation behavior and network variability on fairness–stability trade-offs in multi-client environments.

A. Experimental Scenario

We focused on multi-metric fairness trade-offs and interaction effects, without modifying the underlying adaptation mechanisms. Table II summarizes the main characteristics of the experimental scenario to ensure clarity and self-containment.

B. Overview of the Analytical Approach

The objective of the proposed methodology is to analyze trade-offs between multiple fairness metrics and Quality of Experience (QoE) indicators in multi-client adaptive streaming systems. Rather than optimizing a specific objective, the approach focuses on interpreting how different adaptation strategies influence fairness dimensions under shared network conditions. All analyses are performed offline using execution logs from controlled experimental scenarios, ensuring

TABLE I: Comparison of Related Works from a Fairness Evaluation Perspective

Work	Adaptation Strategy	Fairness Perspective	Evaluation Limitations	Evaluation Scope
[1]	Heuristic ABR (FESTIVE)	Implicit fairness via bandwidth sharing	No explicit fairness metrics; unstable under high heterogeneity	Early analysis of fairness side-effects in ABR
[3]	Heuristic ABR (BOLA)	QoE stability-focused	Fairness not evaluated; single-client oriented	Robust buffer-based bitrate adaptation
[4]	RL-based ABR (Pensieve)	QoE-centric optimization	Fairness not considered during decision-making	Learned bitrate adaptation for individual QoE
[11]	RL-based ABR (Oboe)	Efficiency and stability	No fairness evaluation	Automated tuning of ABR parameters
[12]	Hybrid ABR (HotDASH)	Throughput efficiency	Ignores inter-client fairness	Content-aware bitrate adaptation
[7]	Fairness metric proposal	QoE distribution fairness (QFS)	Metric-level evaluation only; no ABR integration	Definition and validation of QoE fairness metric
[13]	Fairness metric proposal	Bitrate allocation fairness (BFI)	Single-metric fairness evaluation	Bitrate-based fairness assessment
[14]	Distributed bandwidth allocation	QoE-aware fairness	Focused on bandwidth allocation; limited metric diversity	Fairness-driven rate allocation at bottlenecks
[15]	Multi-agent RL-based ABR	Fairness-aware coordination	Limited fairness metrics; scenario-specific evaluation	Coordinated bitrate decisions in multi-user scenarios
[16]	Edge-assisted streaming	QoE improvement at the edge	Fairness treated implicitly or post-hoc	Edge-cloud collaboration for streaming
[19]	MEC-based optimization	QoE fairness	Analytical modeling; limited experimental analysis	Fairness-aware resource allocation in MEC
[20]	Edge-assisted ABR	Fairness via JFI	Single fairness metric; no trade-off analysis	Fairness-constrained ABR at the edge
This Work	**Analytical Study**	**Multi-metric characterization**	**Observational perspective only**	**Characterizing fairness and stability trade-offs**

TABLE II: Summary of Experimental evaluation scenario.

Aspect	Description
Scenario Type	Multi-client adaptive streaming under shared edge
Clients	10 concurrent DASH clients in Docker containers
Streaming Engine	DASH.js player automated via Selenium
Network Traces	Verizon-LTE ($\mu \approx 20$ Mbps) and UMTS ($\mu \approx 2$ Mbps)
Video Content	Big Buck Bunny (6 levels: 360p–2160p)
Session	120 seconds of playback per client

that adaptation decisions remain unaffected by the evaluation process.

A structured analytical workflow is adopted in which executions are grouped into observation sets, complementary fairness metrics are computed independently, inter-metric correlations are evaluated, and adaptation strategies are mapped into a fairness–stability interaction space. This enables systematic identification of metric conflicts and trade-off regions under identical network conditions.

C. Observation Sets and Experimental Grouping

To ensure consistency and interpretability, experimental results are organized into observation sets that group executions sharing comparable network characteristics. Each observation set corresponds to a specific class of network behavior, such as relatively stable conditions or highly variable throughput patterns. By avoiding the aggregation of heterogeneous scenarios, the analysis prevents misleading conclusions caused by uncontrolled variability. Within each observation set, all ABR strategies are evaluated using the same set of clients, network traces, and execution parameters. This grouping enables direct comparison of fairness outcomes and QoE stability across adaptation strategies, highlighting how identical conditions can lead to different fairness profiles.

D. Multi-Metric Fairness Characterization

To move beyond single-index evaluation, we define four fairness indicators that quantify distinct aspects of resource distribution and playback stability:

- **TF-QoE (Time-Fair QoE):** Measures the relative deviation of client i's composite QoE score (Q_i) from the session average ($\bar{Q}$):

$$TF\text{-}QoE_i = 1 - \frac{|Q_i - \bar{Q}|}{\bar{Q} + \epsilon} \tag{1}$$

  where ϵ is a small constant (10^{-6}) for numerical stability.
- **QFS (QoE Fairness Score):** An application of Jain's Fairness Index (JFI) over the temporal average of composite QoE score samples ($\hat{Q}_i$):

$$QFS = \frac{\left(\sum_{i=1}^{N} \hat{Q}_i\right)^2}{N \cdot \sum_{i=1}^{N} \hat{Q}_i^2} \tag{2}$$

- **BFI (Bitrate Fairness Index):** Evaluates fairness in quality level allocation using JFI on average numeric representation indices ($\bar{L}_i$):

$$BFI = \frac{\left(\sum_{i=1}^{N} \bar{L}_i\right)^2}{N \cdot \sum_{i=1}^{N} \bar{L}_i^2} \tag{3}$$

  where $\bar{L}_i$ ranges from 1 (360p) to 5 (2160p).
- **SAF (Stability-Aware Fairness):** Characterizes the equity of adaptation stability by considering the count of quality switches (S_i):

$$SAF = 1 - \frac{|S_i - \bar{S}|}{\bar{S} + \epsilon} \tag{4}$$

This section presents the results of the multi-metric fairness analysis based on previously collected experimental data. The

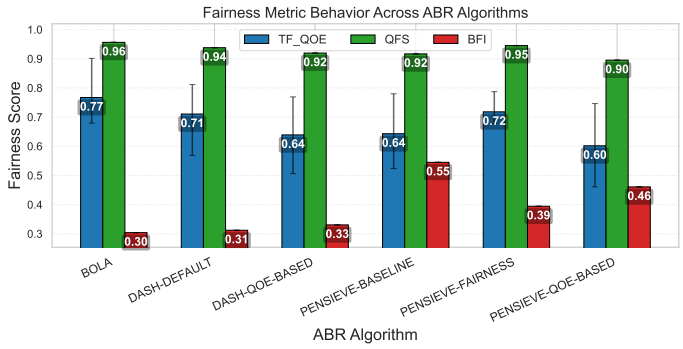

Fig. 1: Fairness metric behavior across adaptive bitrate (ABR) strategies under multi-client conditions, comparing QFS, TF-QoE, and BFI.

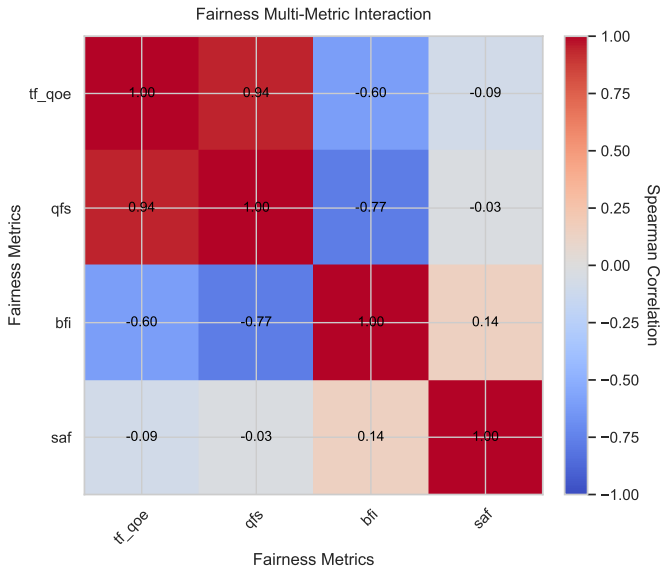

Fig. 2: Multi-metric fairness interaction under multi-client conditions, based on Spearman correlation analysis.

discussion focuses on how different adaptive bitrate (ABR) strategies influence fairness outcomes and QoE stability under shared and constrained network conditions. The results reveal consistent behavioral patterns that differentiate adaptation strategies across fairness and stability dimensions and follow the observation sets defined in Section IV, ensuring fair comparisons under comparable network conditions.

E. Fairness Metric Behavior Across Adaptation Strategies

The analysis reveals that different ABR strategies exhibit distinct fairness profiles depending on the metric considered. Metrics capturing distributional aspects of user experience, such as QoE Fairness Score (QFS), often favor strategies that prioritize stable QoE exposure, whereas metrics related to resource allocation, such as Bitrate Fairness Index (BFI), may highlight different adaptation behaviors. This divergence indicates that fairness improvements observed under one metric may coexist with degradation according to another, reinforcing the need for a multi-metric evaluation perspective.

Figure 1 illustrates this behavior under multi-client conditions. While most ABR strategies achieve consistently high QoE Fairness Score (QFS) values (typically above ≈ 0.9), substantial disparities emerge when temporal fairness (TF-QoE) and bitrate distribution fairness (BFI) are considered. In particular, TF-QoE exhibits noticeable variation across adaptation strategies, whereas BFI shows pronounced degradation for algorithms that prioritize individual QoE maximization, with values dropping to the 0.7–0.8 range in some cases. These results indicate that strategies maintaining stable QoE exposure across users do not necessarily ensure equitable bitrate allocation, especially under heterogeneous network conditions. Consequently, single-metric fairness evaluation may obscure relevant trade-offs, reinforcing the need for a multi-dimensional fairness analysis in adaptive streaming systems.

F. Multi-Metric Fairness Interaction

Figure 2 analyzes the interaction between different fairness metrics under multi-client conditions using Spearman correlation analysis. The results reveal a strong positive correlation between TF-QoE and QoE Fairness Score (QFS) ($\rho \approx 0.94$), indicating that both metrics capture closely aligned aspects of perceived user experience fairness. In contrast, pronounced negative correlations are observed between QoE-oriented fairness metrics and bitrate distribution fairness. Specifically, QFS exhibits a strong negative correlation with the Bitrate Fairness Index (BFI) ($\rho \approx -0.77$), while TF-QoE shows a moderate negative correlation with BFI ($\rho = -0.60$). These results indicate that improvements in perceived QoE fairness are often accompanied by increased imbalance in bitrate allocation among competing clients. Stability-Aware Fairness (SAF), in turn, presents only weak correlations with the remaining metrics ($|\rho| \leq 0.14$), suggesting that adaptation stability captures a largely orthogonal fairness dimension. Overall, these interactions demonstrate that fairness metrics do not evolve independently and may encode competing objectives, particularly in heterogeneous and congested environments. Consequently, aggregated or single-metric fairness evaluation may obscure underlying conflicts between fairness dimensions, reinforcing the need for multi-metric analysis when interpreting adaptive streaming behavior.

G. Trade-offs Between Fairness and QoE Stability

A recurring pattern observed across scenarios is the interaction between fairness outcomes and QoE stability indicators, such as quality oscillations and rebuffering behavior. Figure 3 illustrates this interaction by jointly analyzing temporal fairness (TF-QoE) and playback stability (SAF) under multi-client conditions. The results show that adaptation strategies occupy distinct regions of the fairness–stability space, indicating that improvements along one dimension do not necessarily translate into gains along the other. Adaptation strategies that aggressively react to short-term throughput variations

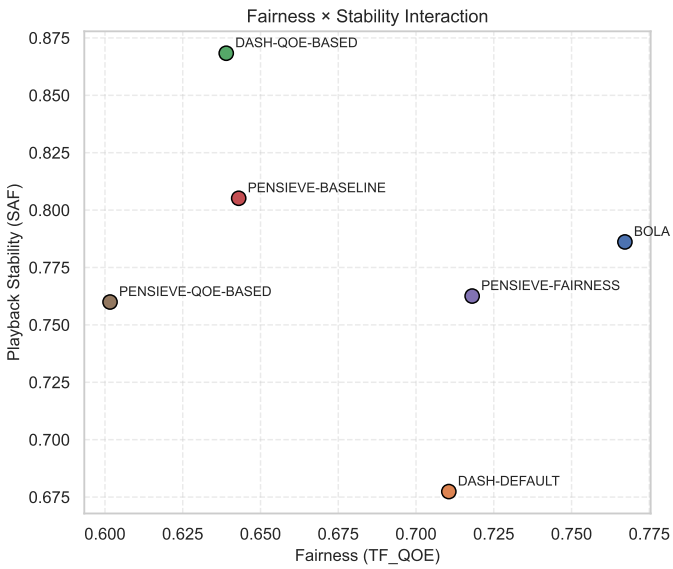

Fig. 3: Fairness versus playback stability interaction under multi-client conditions, relating TF-QoE and SAF across ABR strategies.

may improve fairness-related metrics but often at the cost of increased playback instability. As illustrated in Figure 3, DASH-DEFAULT achieves moderate temporal fairness (TF-QoE ≈ 0.71) but exhibits the lowest playback stability among the evaluated strategies (SAF ≈ 0.68). In contrast, DASH-QoE-BASED prioritizes stability, reaching the highest SAF value (SAF ≈ 0.87), while operating at lower temporal fairness levels (TF-QoE ≈ 0.64). RL-based strategies occupy intermediate regions of the fairness–stability space. For instance, PENSIEVE-FAIRNESS improves temporal fairness relative to the baseline (TF-QoE ≈ 0.72) while maintaining moderate stability (SAF ≈ 0.76), whereas BOLA simultaneously achieves high fairness and stability (TF-QoE ≈ 0.77, SAF ≈ 0.79). These results confirm that fairness and stability objectives are not always aligned and that their trade-offs become more pronounced under heterogeneous and variable network conditions.

H. Impact of Network Variability on Fairness Trade-offs

Network variability acts as a catalyst for fairness degradation, specifically by triggering asymmetric adaptation cycles. Figure 4 illustrates the behavior under the Verizon-LTE trace. During throughput dips (e.g., the 40s mark), reactive strategies exhibit a "tactical panic" where standard deviation spikes exceed 10 Mbps. This indicates that while the total bandwidth is equally reduced, the *rate* of quality downswitching varies significantly across clients due to slightly different buffer pressures. Furthermore, we identify a "recovery lag" phenomenon: as bandwidth stabilizes after the 80s mark, some clients sustain lower-quality levels longer than necessary to replenish buffers, creating a persistent gap in TF-QoE. This behavioral heterogeneity remains obscured in distributional

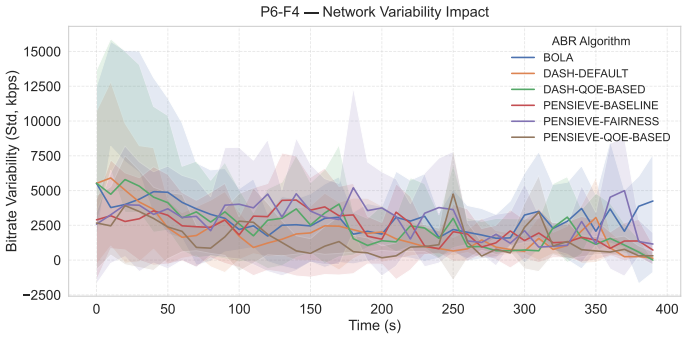

Fig. 4: Network variability impact on bitrate adaptation over time under multi-client conditions. The figure shows standard deviation of bitrate selection across 10 clients replaying the Verizon-LTE trace.

metrics like QFS or JFI but proves critical for understanding the structural unfairness inherent in decentralized ABR logic.

I. Implications for Fairness-Aware ABR Design

Our results suggest that achieving holistic fairness requires moving beyond individual bitrate maximization. We propose three actionable design shifts:

- **Inter-Client Stability Constraints**: ABR controllers should incorporate the variance of switch counts $\sigma(S)$ into their objective functions. Designers should penalize strategies where specific users suffer disproportionate quality oscillations compared to the fleet average (SAF-aware control).
- **Synchronized Recovery Windows**: To mitigate the TF-QoE "recovery lag," edge-assisted controllers should utilize cross-client synchronization to coordinate quality upswitches, preventing a subset of clients from monopolizing throughput during bandwidth recovery.
- **Asymmetric Reward Penalties**: In cellular-dominant environments, fairness-aware RL reward functions (R) should prioritize the standard deviation of perceived QoE over the session average ($\bar{Q}$), explicitly treating fairness as a stability-linked temporal dimension.

J. Discussion and Implications

From an analytical perspective, the results show that fairness trade-offs are structural properties emerging from the interaction between adaptation logic and network variability, rather than anomalies of specific algorithms. Fairness in multi-client adaptive streaming is inherently multi-dimensional, and no single ABR strategy consistently dominates across all fairness metrics and QoE stability indicators. Instead, strategies occupy distinct regions of the fairness–stability space, reflecting implicit trade-offs between responsiveness and conservatism. Improvements in distributional fairness, bitrate allocation, or temporal QoE exposure do not necessarily occur simultaneously. Aggressive adaptation may enhance certain fairness

metrics while degrading playback stability, whereas conservative strategies preserve stability at the cost of increased long-term disparities. Network variability amplifies these effects: under stable conditions, fairness outcomes tend to converge, while highly variable environments expose latent conflicts between fairness dimensions. These findings highlight the limitations of single-metric evaluation and carry direct implications for fairness-aware ABR design. Single-metric optimization is structurally insufficient in multi-client environments, and adaptation mechanisms—particularly reinforcement learning policies—should incorporate multi-dimensional evaluation criteria that explicitly account for network variability as a fairness amplifier.

V. Conclusion and Future Work

This study demonstrates that fairness trade-offs are structural characteristics of multi-client adaptive streaming systems. Through a multi-metric observational analysis, we show that fairness is inherently multi-dimensional and cannot be adequately captured by single or aggregated indicators. Different ABR strategies occupy distinct regions of the fairness–stability space, reflecting implicit trade-offs between responsiveness and conservatism under shared and heterogeneous network conditions. The results further indicate that network variability acts as a fairness amplifier, exposing latent conflicts between fairness dimensions that remain obscured under stable scenarios. Rather than identifying a universally optimal strategy, this work emphasizes the need for structured, multi-dimensional fairness evaluation frameworks that explicitly account for interacting fairness metrics and network dynamics when assessing adaptive streaming behavior. Future work will investigate how these analytical insights can inform the design of fairness-aware adaptation mechanisms and multi-objective learning strategies capable of balancing competing fairness dimensions in dynamic edge-assisted environments.

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